4.Implement histogram equalization on the given image and compare it with the original image using Open CV.

code;

```
import cv2
import matplotlib.pyplot as plt

# Read the image
img = cv2.imread(r"C:\Users\ROHIT\Pictures\Instagram\IMG_20230206_213414_457.jpg")

if img is None:
    print("Image not found!")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply histogram equalization
    equalized = cv2.equalizeHist(gray)

    # Display images
    plt.subplot(1, 2, 1)
    plt.imshow(gray, cmap='gray')
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1, 2, 2)
    plt.imshow(equalized, cmap='gray')
    plt.title("Equalized Image")
    plt.axis("off")

    plt.show()
```

output

